BME i9400 — Fall 2026

Practice midterm examination

For the review session · 75 minutes · Meetings 1–12

Name: _____

Instructions. This practice exam has the same format, length, and rules as the midterm on Tuesday, November 3. Closed book and closed notes. A basic or scientific calculator is permitted; phones, laptops, and other devices are not. A reference sheet is provided on the next page. Show your work on the problems. Answers may be left as fractions.

Part	Content	Points
A	12 multiple-choice questions (A1–A11: 2 points each; A12: 3 points)	25
B	Problems: answer **any three of the four**, 25 points each	75
	Total	**100**

Part B: circle the three problems you want graded. B1 B2 B3 B4

Reference sheet

The confusion matrix

	predicted positive	predicted negative
actually positive	TP (true positive)	FN (false negative)
actually negative	FP (false positive)	TN (true negative)

Quantity	Formula
sensitivity (recall, true positive rate)	$TP/(TP+FN)$
specificity (true negative rate)	$TN/(TN+FP)$
false positive rate	$FP/(FP+TN) = 1 - \text{specificity}$
positive predictive value (PPV, precision)	$TP/(TP+FP)$
negative predictive value (NPV)	$TN/(TN+FN)$
accuracy	$(TP+TN)/n$
positive likelihood ratio	$\text{LR}^+ = \text{sensitivity}/(1-\text{specificity})$
negative likelihood ratio	$\text{LR}^- = (1-\text{sensitivity})/\text{specificity}$

Probability and odds

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} \qquad \text{odds} = \frac{p}{1-p}, \quad p = \frac{\text{odds}}{1+\text{odds}} \qquad \text{posterior odds} = \text{LR} \times \text{prior odds}$$

Models and losses

$$\sigma(z) = \frac{1}{1+e^{-z}} \qquad \text{cross-entropy for one sample: } -\left[\, y \ln p + (1-y)\ln(1-p) \,\right]$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2} \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \qquad C = \frac{1}{n-1}\tilde{X}^\top \tilde{X}$$

Part A — Multiple choice

Circle the **one** best answer. A1–A11 are worth 2 points each; A12 is worth 3 points.

A1. A disease has prevalence 10%. A test has sensitivity 0.80 and specificity 0.90. Approximately what fraction of people who test positive have the disease?

(a) 0.80 (b) 0.47 (c) 0.17 (d) 0.10

A2. A test has sensitivity 0.90 and specificity 0.95. Its positive likelihood ratio, $\mathrm{LR}^{+} = \text{sensitivity}/(1-\text{specificity})$, is

(a) 0.95 (b) 1.9 (c) 9 (d) 18

A3. Which pair of vectors is orthogonal?

(a) $(1, 2)$ and $(2, 1)$ (b) $(1, 1)$ and $(2, 2)$ (c) $(1, 2)$ and $(-2, 1)$ (d) $(1, 0)$ and $(1, 1)$

A4. $\tilde{X}$ is a centered data matrix with n samples (rows) and d features (columns). The covariance matrix $C = \frac{1}{n-1}\tilde{X}^{\top}\tilde{X}$ has shape

(a) $n \times n$ (b) $d \times d$ (c) $n \times d$ (d) $d \times 1$

A5. The matrix $X^{\top}X$ in the normal equations fails to be invertible when

(a) there are more samples than features and no feature is a combination of the others

(b) the features have been standardised

(c) there are fewer samples than weights, or one feature is an exact linear combination of others

(d) the target is noisy

A6. In ridge regression with an unpenalised intercept, as $\lambda \to \infty$ every prediction approaches

(a) the mean of the training targets

(b) the least squares prediction

(c) zero

(d) the largest training target

A7. At the threshold $p = 0.5$, the decision boundary of a logistic regression with features x and weights w is

(a) the set of points where $x^\top w = 0$: a line, plane, or hyperplane

(b) always a circle centred on the mean of the data

(c) the set of points where $p = 0$

(d) determined by the prevalence alone

A8. On a test set with prevalence 10%, a classifier that predicts "negative" for everyone has accuracy

(a) 0.10 (b) 0.50 (c) 0.90 (d) undefined

A9. In 5-fold cross-validation, each sample is

(a) used for training exactly once

(b) never used for training

(c) used for evaluation in every fold

(d) used for evaluation exactly once and for training four times

A10. A model assigns a probability of 0.8 to 50 women, of whom 30 develop diabetes. For this group of women, the model

(a) is calibrated

(b) overestimates the risk

(c) underestimates the risk

(d) has an AUC of 0.6

A11. A classifier for a condition with prevalence 0.5% has an AUC of 0.95. Which additional quantity best tells a clinician what fraction of the patients it flags actually have the condition?

(a) the precision (PPV) at the threshold that will be used

(b) the specificity

(c) the AUC on a second test set

(d) the sensitivity

A12. *(3 points)* Cross-validation is run for 20 values of a hyperparameter, and the best of the 20 cross-validated scores is reported as the model's performance. Which procedure gives an unbiased estimate instead?

(a) report the average of the 20 scores

(b) repeat the cross-validation with a different random partition and report the best score again

(c) refit the best model to all of the data and report its training score

(d) nested cross-validation: choose the hyperparameter inside an inner loop, and score the result on outer folds that the choice never saw

Part B — Problems (25 points each; answer any three)

B1. Screening

A hypothetical blood test for prostate cancer has sensitivity 0.90 and specificity 0.80. It is used to screen 1,000 men, of whom 5% have the cancer.

(a) (6 pts) Fill in the table of expected counts.

	test positive	test negative	total
cancer			
no cancer			
total			1,000

(b) (5 pts) Compute the PPV and the NPV.

(c) (7 pts) Compute the likelihood ratios $\mathrm{LR}^+ = \text{sensitivity}/(1 - \text{specificity})$ and $\mathrm{LR}^- = (1 - \text{sensitivity})/\text{specificity}$. Using posterior odds = LR$\times$ prior odds, compute the probability of cancer for a man who tests **negative**, and check it against the table.

(d) (7 pts) Every man who tests positive is given a second, independent test with the same sensitivity and specificity. What is the probability of cancer for a man who tests positive twice?

B2. Vectors and principal components

(a) (6 pts) For the two patients $x = (3, 4)$ and $z = (4, 3)$, compute $\|x\|$, the distance $\|x - z\|$, and $\cos\theta$ between them.

(b) (4 pts) For the unit vector $u = (0.8, 0.6)$, compute the coordinate of x along u and the projection of x onto u.

(c) (9 pts) Four patients have the centered data matrix $\tilde{X} = \begin{bmatrix} 3 & 0 \\ -3 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}$. Compute the covariance matrix, state the two principal components and their variances, and give the fraction of variance explained by the first.

(d) (6 pts) Each feature is now standardised (divided by its standard deviation). What are the covariance matrix and the fraction of variance explained by the first component after standardisation? What does this say about applying PCA to features measured in different units?

B3. Linear regression

Four patients have one feature x and a target y:

x	0	1	2	3
y	1	3	5	5

The model is $\hat{y} = w_0 + w_1 x$.

(a) (4 pts) Write the design matrix X (with a column of ones) and the vector y.

(b) (5 pts) Compute $X^\top X$ and $X^\top y$.

(c) (7 pts) Solve the normal equations $X^\top X\, w = X^\top y$. For a 2×2 matrix, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

(d) (6 pts) Compute the residuals, verify that $X^\top r = 0$, and compute the RMSE.

(e) (3 pts) Compute $R^2 = 1 - \dfrac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$, where $\bar{y}$ is the mean of y.

B4. Logistic regression and evaluation

(a) (9 pts) A logistic regression for diabetes has log-odds $z = -1.1 + 0.69\,g$, where g is standardised glucose. Use $e^{-1.1} \approx 1/3$ and $e^{0.69} \approx 2$ to compute the odds and the probability of diabetes for $g = 0$, $g = 1$, and $g = 2$.

Another model assigns probabilities to ten held-out women:

p	0.2	0.2	0.2	0.2	0.2	0.8	0.8	0.8	0.8	0.8
y	1	0	0	0	0	1	1	0	1	0

(b) (9 pts) At the threshold 0.5, give the confusion matrix and compute the sensitivity, specificity, precision, and accuracy.

(c) (7 pts) For each of the two probability values, compare the predicted probability with the observed fraction of women with diabetes. Are the model's probabilities trustworthy as risks?

End of examination